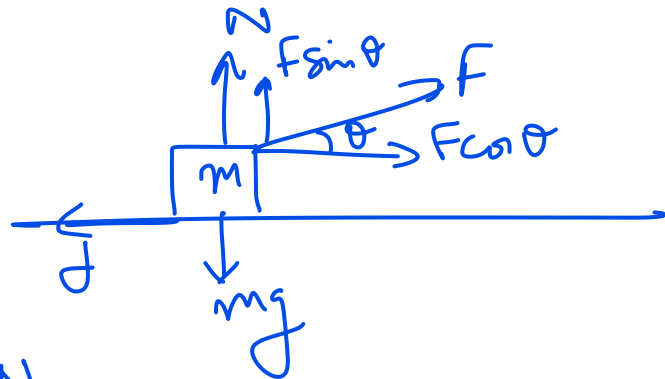


1. Variable Force

A block of mass m is pulled along a horizontal surface by applying a force at an angle θ with the horizontal. If the block travels with a uniform velocity and has a displacement d , and the coefficient of friction is μ , then the work done by the applied force is:

- $\frac{\mu mgd}{\cos \theta + \mu \sin \theta}$
- $\frac{\mu mgd \cos \theta}{\cos \theta + \mu \sin \theta}$
- $\frac{\mu mgd \sin \theta}{\cos \theta + \mu \sin \theta}$
- $\frac{\mu mgd \cos \theta}{\cos \theta - \mu \sin \theta}$



$$\Rightarrow f = \mu N$$

$$\Rightarrow N + F \sin \theta = mg$$

$$\Rightarrow f = \mu (mg - F \sin \theta)$$

$$\Rightarrow F \cos \theta - f = m \cdot a = m(0) = 0$$

$$\Rightarrow F \cos \theta - \mu (mg - F \sin \theta) = 0$$

$$\Rightarrow F (\cos \theta + \mu \sin \theta) = \mu mg$$

$$\Rightarrow F = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$$

$$\Rightarrow W = F \cdot d = \frac{\mu mgd}{\cos \theta + \mu \sin \theta}$$

$\Downarrow$
Option - A